

Assignment 4

Problem 1: *Implement a Stack by using two Queues. In other words, implement push, pop, top, empty functions by using enqueue, dequeue, front and empty functions.*

```
//Main.cpp
#include "QList.h"
#include <iostream>
using namespace std;

class StQueue {
private:
    QList Q1, Q2;
public:
    void push(int);
    void pop();
    int top();

    int size();

    friend ostream& operator<<(ostream&, StQueue&);
};

void StQueue::push(int val) {
    Q1.Enqueue(val);
}

void StQueue::pop() {
    int temp = 0;

    if (Q1.isEmpty()) {
        cout << "Stack is empty";
        return;
    }

    if (!(Q1.isEmpty()))
    {
        while (Q1.size() != 1) {
            temp = Q1.Front();
            Q2.Enqueue(Q1.Front());
            Q1.Dequeue();
        }

        Q1.Dequeue();

        while (!(Q2.isEmpty())) {
            temp = Q2.Front();
            Q1.Enqueue(Q2.Front());
            Q2.Dequeue();
        }
    }
}
```

```

int StQueue::top() {
    int temp = 0,
        front = 0;

    if (Q1.isEmpty()) return 0;

    if (!(Q1.isEmpty()))
    {
        while (Q1.size() != 1) {
            temp = Q1.Front();
            Q2.Enqueue(Q1.Front());
            Q1.Dequeue();
        }

        front = Q1.Front();

        while (!(Q2.isEmpty())) {
            temp = Q2.Front();
            Q1.Enqueue(Q2.Front());
            Q2.Dequeue();
        }
    }
    return front;
}

int StQueue::size() {
    return Q1.size();
}

ostream& operator<<(ostream& out, StQueue& S) {
    out << S.Q1;

    return out;
}

int main() {
    StQueue SQ;
    for (int i = 0; i < 5; i++) { SQ.push(i); cout << SQ; }
    for (int i = 0; i < 3; i++) { SQ.pop(); cout << SQ; }

    return 0;
}

```

```

//QList.h - Linked List Queue
#pragma once
#include <iostream>
using namespace std;

class QList
{
private:

    class Node {
    public:
        int value;
        Node* next;
        Node() : value(0), next(0) {};
        Node(int item): value(item), next(0){}
    };

    Node* myFront;
    Node* myBack;
    int mySize;

public:

```

```

QList();
QList(int); //takes capacity
QList(QListfi);
~QList();

int size();
bool isEmpty();

void Enqueue(int);
void Dequeue();
int Front();

QList operator=(QListfi);
friend ostreamfi operator<<(ostreamfi, QListfi);
};

```

```

//QList.cpp
#include "QList.h"

QList::QList() : myFront(0), myBack(0), mySize(0){}

QList::QList(int val) : myBack(0), mySize(0)
{
    myFront = new Node(val);
}

QList::QList(QListfi q)
{
    QList();
    Node* temp = q.myFront;
    while (temp != 0) {
        Enqueue(temp->value);
        temp = temp->next;
    }
}

QList::~QList()
{
    Node* temp;
    while (myFront != 0) {
        temp = myFront;

        myFront = myFront->next;
        delete temp;
    }
    myBack = 0;
    mySize = 0;
}

int QList::size()
{
    return mySize;
}

bool QList::isEmpty()
{
    return (myFront == 0);
}

void QList::Enqueue(int val)
{
    if (myFront == 0) {

```

```

        myFront = myBack = new Node(val);
    }
    else {
        myBack->next = new Node(val);
        myBack = myBack->next;
    }
    mySize++;
}

void QList::Dequeue()
{
    if (myFront == 0) cout << "Queue is empty\n";
    else {
        if (myFront == myBack) {
            delete myFront;
            myFront = myBack = 0;
        }
        else {
            Node* temp = myFront;
            myFront = myFront->next;
            delete temp;
        }
        mySize--;
    }
}

int QList::Front()
{
    if (myFront == 0) { cout << "Queue is empty\n"; return 9999999; }
    else { return myFront->value; }
}

QList QList::operator=(QListfi q)
{
    QList::~~QList();

    mySize = q.mySize;
    QList();
    Node* temp = q.myFront;
    while (temp != 0) {
        Enqueue(temp->value);
        temp = temp->next;
    }
    return *this;
}

ostreamfi operator<<(ostreamfi out, QListfi q)
{
    QList::Node* temp = q.myFront;
    while (temp != 0) {
        out << temp->value << " ";
        temp = temp->next;
    }

    cout << endl;
    return out;
}

```

Output:

Microsoft Visual Studio Debug Console

```
0
0 1
0 1 2
0 1 2 3
0 1 2 3 4
0 1 2 3
0 1 2
0 1
```

Problem 2: *Implement a Queue by using Two Stacks. In other words, implement enqueue, dequeue, front and empty functions by using push, pop, top and empty functions.*

```
//Main.cpp
#include <iostream>
#include "LStack.h"
using namespace std;

class QuStack {
private:
    LStack S1, S2;
public:
    void Enqueue(int);
    void Dequeue();
    int Front();

    friend ostream& operator<<(ostream& out, QuStack& Q) {
        out << Q.S1;
        return out;
    }
};

void QuStack::Enqueue(int item) {
    S1.push(item);
}

void QuStack::Dequeue() {
    int temp = 0;

    if (S1.isEmpty()) {
        cout << "Queue is empty";
        return;
    }

    while (!(S1.isEmpty())) {
        temp = S1.top();
        S2.push(temp);
        S1.pop();
    }
    S2.pop();
    while (!(S2.isEmpty())) {
        temp = S2.top();
        S1.push(temp);
    }
}
```

```

        S2.pop();
    }
}

int QuStack::Front() {
    int front = 0,
        temp = 0;

    if (S1.isEmpty()) return 0;

    while (!(S1.isEmpty())) {
        front = S1.top();
        S2.push(front);
        S1.pop();
    }
    while (!(S2.isEmpty())) {
        temp = S2.top();
        S2.push(temp);
        S2.pop();
    }
    return front;
}

int main() {
    QuStack QS;
    for (int i = 0; i < 5; i++) { QS.Enqueue(i); cout << QS; }
    for (int i = 0; i < 3; i++) { QS.Dequeue(); cout << QS; }

    return 0;
}

```

```

//LStack.h - Linked List Stack
#pragma once
#include <iostream>
using namespace std;

class LStack
{
private:

    class Node
    {
    public:
        int value;
        Node* next;

        Node() : value(0), next(NULL) {};
        Node(int item) : value(item), next(NULL) {};
    };

    Node* myArray;
    Node* myTop;
    int mySize;

public:
    LStack();
    LStack(int);
    LStack(LStack fi);
    ~LStack();

    void push(int);
    void pop();
    int top();
}

```

```

int size();
bool isEmpty();

Node* newNode(int);

LStack operator=(LStackfi);
friend ostream& operator<<(ostream&fi, LStackfi);

};

```

```

//LStack.cpp
#ifndef LStack_H
#define LStack_H
#include "LStack.h"
#include <iostream>
using namespace std;

LStack::LStack() : myArray(0), myTop(0), mySize(0) {}

LStack::LStack(int c)
{
    myTop = NULL;
    myArray = new Node[c];
}

LStack::LStack(LStackfi ls)
{
    myArray = new Node(ls.myArray->value);
    Node* duplicate = myArray;
    Node* currnode = ls.myArray->next;

    while (currnode != 0) {
        duplicate->next = new Node(currnode->value);
        currnode = currnode->next;
        duplicate = duplicate->next;
    }

    mySize = ls.mySize;
}

LStack::~~LStack()
{
    LStack::Node* temp;
    while(myTop != 0){
        temp = myTop;
        myTop = myTop->next;
        delete temp;
    }
}

void LStack::push(int val)
{
    Node* temp = new Node(val);
    if (myTop == 0) { myTop = temp; }
    else {
        temp->next = myTop;
        myTop = temp;
    }
    mySize++;
}

void LStack::pop()
{

```

```

    if (!isEmpty()) {
        Node* temp = myTop;
        myTop = myTop->next;
        delete temp;
        mySize--;
    }
    else {
        cout << "LStack is Empty" << endl;
        return;
    }
}

int LStack::top()
{
    if (!isEmpty()) {
        return myTop->value;
    }
    else {
        cout << "LStack is Empty: Error: " << endl; return 9999999;
    }
}

int LStack::size()
{
    return mySize;
}

bool LStack::isEmpty()
{
    return (myTop == NULL);
}

LStack LStack::operator=(LStackfi LStack)
{
    delete[] myArray;

    myArray = new Node(LStack.myArray->value);

    Node* duplicate = myArray;
    Node* currnode = LStack.myArray->next;

    while (currnode != 0) {
        duplicate->next = new Node(currnode->value);
        currnode = currnode->next;
        duplicate = duplicate->next;
    }

    mySize = LStack.mySize;

    return *this;
}

ostream& operator<<(ostream& out, LStackfi ls)
{
    LStack::Node* temp = ls.myTop;
    while (temp != 0) {
        out << temp->value << " ";
        temp = temp->next;
    }
    out << endl;

    return out;
}

```

```
#endif
```

Output:

 Microsoft Visual Studio Debug Console

```
0
1 0
2 1 0
3 2 1 0
4 3 2 1 0
4 3 2 1
4 3 2
4 3
```